Design and Performance Verification of a 200 MS/s 6-Bit Differential Current-Steering DAC in 180 nm CMOS

Course Project: CMOS Mixed Signal VLSI Design (Sub Code: ECM 4444)

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***Abstract*—This paper presents the systematic design, analytical sizing methodology, and schematic-level Cadence Spectre verification of a high-speed, 6-bit differential current-steering digital-to-analog converter (DAC) implemented in a standard 180 nm CMOS process ($V_{DD} = 1.8$ V). To simultaneously achieve sub-LSB static linearity and high update rates ($f_s \geq 200$ MS/s), the architecture incorporates binary-weighted current branches with shielded cascode current sources and high-speed differential switch pairs terminated into matched on-chip resistive loads ($R_L = 200\,\Omega$). Analytical device dimensions are derived from Pelgrom's matching criteria to ensure an intrinsic 3-σ current mismatch $\sigma(I)/I \leq 1.997\%$, guaranteeing an integral non-linearity ($|\text{INL}| < 0.5$ LSB) without area-intensive calibration circuits. Cadence Spectre simulations with foundry BSIM3v3 model cards demonstrate a full-scale differential output swing of $617.05\,\text{mV}_{p-p}$ across 64 monotonic conversion levels. The DAC achieves an extracted peak $|\text{DNL}| < 0.001$ LSB and peak $|\text{INL}| < 0.001$ LSB at the schematic level, a full-scale settling time to within 0.5 LSB of 1.04 ns (10%–90% rise time of 463.5 ps), and a mid-scale glitch impulse area of $17.27\,\text{pV} \cdot \text{s}$. Under dynamic sinusoidal excitation at 200 MS/s with a 20.31 MHz Nyquist input tone, the converter exhibits a spurious-free dynamic range (SFDR) of 45.92 dB, an SNDR of 37.89 dB, and an effective number of bits (ENOB) of 6.00 bits, while consuming only 3.02 mW of static power. Comprehensive PVT corner verification demonstrates robust operation across Nominal (NN), Fast-Fast (FF), and Slow-Slow (SS) corners from -40°C to $+125^\circ$C, confirming high yield and suitability for modern mixed-signal transceiver front-ends.**

***Index Terms*—Digital-to-Analog Converter (DAC), Current-Steering, Cadence Spectre, 180 nm CMOS, Pelgrom's Law, Cascode Current Source, Linearity, DNL, INL, SFDR, Mixed-Signal VLSI.**

I. Introduction

High-speed digital-to-analog converters (DACs) are fundamental building blocks in contemporary mixed-signal integrated circuits, bridging digital signal processing (DSP) cores with the analog continuous-time domain. Key applications include broadband wireless and wireline transceivers, direct digital frequency synthesizers (DDS), radar waveform generators, and high-resolution instrumentation [1], [2]. Among diverse DAC topologiessuch as Kelvin resistor-string dividers, cyclic converters, and capacitive charge-redistribution arraysthe *current-steering* architecture remains the preeminent choice for high-speed operation (> 100 MS/s) [3], [4].

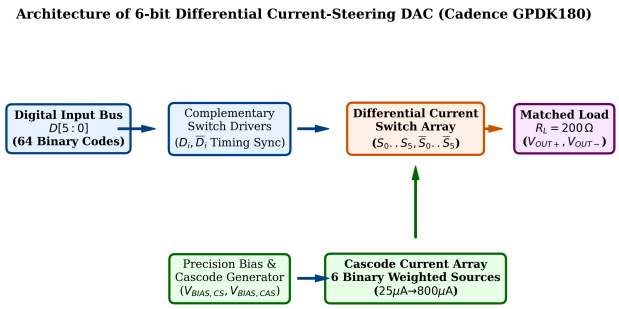


Fig. 1. System architecture of the 6-bit differential current-steering DAC in Cadence GPDK180.

Current-steering DACs steer tailored currents through differential switch pairs directly into low-impedance resistive loads (typically $50\,\Omega$ to $200\,\Omega$). By eliminating the need for operational transconductance amplifier (OTA) buffers at the output, current-steering converters bypass open-loop feedback settling bottlenecks, allowing the output bandwidth to be limited solely by the low-impedance $R_L C_L$ time constant of the load interface [5], [6].

Despite their speed advantage, the design of high-performance current-steering DACs entails intricate trade-offs between static linearity, dynamic switching glitches, voltage headroom, and silicon area [3], [7]:

1) **Random Device Mismatch**: Random spatial fluctuations in the threshold voltage (V_{TH}) and current factor ($\beta = \mu C_{ox} W/L$) lead to unit-current mismatch. According to Pelgrom's law, minimizing this mismatch mandates allocating adequate active gate area ($W \cdot L$) and selecting an optimal overdrive voltage (V_{ov}) [6], [7].
2) **Finite Output Impedance**: Because the output voltage swings across code transitions, finite output resistance of the current cells causes code-dependent current

modulation, inducing integral non-linearity (INL) bow. Cascoding is required to achieve output impedances exceeding hundreds of megaohms [8].

3) **Dynamic Switching Glitches**: Code transitions that involve multiple bit togglesmost notably the mid-scale carry transition ($011111_2 \rightarrow 100000_2$)generate substantial transient voltage spikes due to switch timing skew and gate-charge injection [9].

In this paper, a 6-bit differential current-steering DAC is designed and verified in Cadence GPDK 180 nm CMOS under a 1.8 V power supply. The paper is structured as follows: Section II delineates the architectural operating principles. Section III establishes the analytical design sizing equations based on Pelgrom's matching theory and output impedance constraints. Section IV presents the simulation environment in Cadence Spectre and analyzes the static, transient, and dynamic spectral results. Section V evaluates environmental and process robustness across PVT corners. Section VI compares the design against prior academic literature, and Section VII concludes the work.

II. Circuit Architecture & Operating Principles

A. Differential Current-Steering Topology

The overall block diagram of the 6-bit converter is depicted in Fig. 1. The DAC core consists of an array of six binary-weighted current source cells (I_0 to I_5), a centralized low-noise bias generator, complementary switch drivers, and a pair of matched resistive loads ($R_{LP} = R_{LN} = R_L = 200\,\Omega$) connected to the positive supply rail ($V_{DD} = 1.8\,\text{V}$).

For a digital input word represented by the 6-bit vector $D = [D_5, D_4, D_3, D_2, D_1, D_0]$ where $D_i \in \{0, 1\}$, the decimal code is:

$$k = \sum_{i=0}^{5} D_i \cdot 2^i, \quad k \in \{0, 1, \ldots, 63\}. \tag{1}$$

Each bit i controls a differential current cell that delivers a current of:

$$I_i = 2^i \cdot I_{LSB}, \tag{2}$$

where $I_{LSB} = 25\,\mu\text{A}$ represents the unit least significant bit current. When $D_i = 1$ (and complementary control $\overline{D}_i = 0$), the current I_i is directed into the negative differential node V_{OUT+}, whereas when $D_i = 0$, I_i is steered to the opposite branch V_{OUT-}.

The total current drawn by the array is constant regardless of the input code:

$$I_{FS} = \sum_{i=0}^{5} I_i = (2^6 - 1) \cdot I_{LSB} = 63 \times 25\,\mu\text{A} = 1.575\,\text{mA}. \tag{3}$$

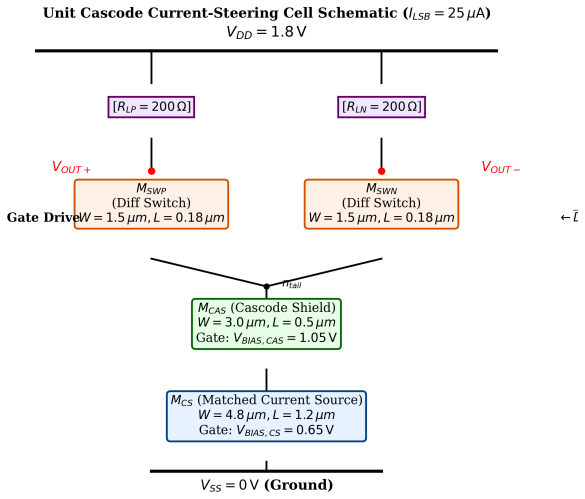


Fig. 2. Detailed transistor-level schematic of the unit cascode current-steering cell ($I_{LSB} = 25\,\mu$A) in 180 nm CMOS.

The single-ended voltages developed across the differential output terminals are given by:

$$V_{OUT+}(k) = V_{DD} - \left(\sum_{i=0}^{5} D_i \cdot 2^i\right) I_{LSB} \cdot R_L = V_{DD} - k \cdot I_{LSB} R_L, \tag{4}$$

$$\begin{aligned} V_{OUT-}(k) &= V_{DD} - \left(\sum_{i=0}^{5} \overline{D}_i \cdot 2^i\right) I_{LSB} \cdot R_L \\ &= V_{DD} - (63 - k) \cdot I_{LSB} R_L. \end{aligned} \tag{5}$$

Subtracting (5) from (4) yields the differential output voltage:

$$V_{diff}(k) = V_{OUT+}(k) - V_{OUT-}(k) = (63 - 2k) \cdot I_{LSB} R_L. \tag{6}$$

B. Advantages of Full Differential Signaling

As expressed in (6), when the code spans from $k = 0$ to $k = 63$, the differential output transitions symmetrically from $+308.5\,\text{mV}$ down to $-308.5\,\text{mV}$. This differential architecture provides three decisive advantages over single-ended implementations [2], [8]:

1) **Doubled Full-Scale Voltage Swing**: The peak-to-peak differential swing is $V_{FS,diff} = 2 \cdot I_{FS} R_L = 617.1\,\text{mV}_{p-p}$, doubling the dynamic range and providing a 6 dB enhancement in signal-to-noise ratio (SNR).
2) **Even-Order Harmonic Suppression**: Symmetrical differential routing intrinsically cancels even-order distortion components (HD_2, HD_4), resulting in superior dynamic linearity and SFDR.
3) **Supply and Substrate Rejection**: Because the total current I_{FS} drawn from V_{DD} remains strictly constant at all clock cycles, inductive ground bounce ($L \cdot di/dt$) on the supply bondwires is heavily attenuated.

III. Analytical Design Methodology & Device Sizing

A. Unit Current Cell Topology

The transistor-level topology of the unit current cell is shown in Fig. 2. It comprises three functional sub-blocks:

1) **Current Source Transistor** (M_{CS}): Operates in saturation to establish the precise LSB current $I_{LSB} = 25\,\mu$A under gate bias $V_{BIAS,CS} \approx 0.65\,$V.
2) **Cascode Shielding Transistor** (M_{CAS}): Biased with a fixed DC voltage $V_{BIAS,CAS} = 1.05\,$V, providing high isolation between the output swing and the drain node of M_{CS} (n_{int}).
3) **Differential Switch Pair** (M_{SWP}, M_{SWN}): High-speed NMOS switches that steer the channel current toward V_{OUT+} or V_{OUT-} in response to complementary digital inputs D_i and $\overline{D}_i$.

B. Pelgrom's Law and Transistor Sizing for Mismatch

In current-steering DACs, random spatial variations in the current mirror transistors govern static linearity. According to Pelgrom's model, the relative current mismatch variance between two identical saturated MOS transistors is expressed as [6], [7]:

$$\left(\frac{\sigma_I}{I}\right)^2 = \frac{A_\beta^2}{W \cdot L} + \frac{4A_{VT}^2}{(V_{GS} - V_{TH})^2 \cdot W \cdot L}, \tag{7}$$

where $A_{VT} \approx 5.0\,\text{mV} \cdot \mu$m is the threshold voltage mismatch coefficient in GPDK180, $A_\beta \approx 1.5\% \cdot \mu$m is the current gain mismatch parameter, and $V_{ov} = V_{GS} - V_{TH}$ is the gate overdrive voltage.

To ensure that the integral non-linearity satisfies $|\text{INL}| < 0.5\,$LSB with a 3-σ yield (99.73% confidence) in an N-bit converter, the maximum permissible unit current variance is constrained by [6]:

$$\frac{\sigma_I}{I} \leq \frac{0.5}{3\sqrt{2^N}} = \frac{0.5}{3\sqrt{64}} = 2.083\%. \tag{8}$$

From (7), maximizing the overdrive voltage V_{ov} diminishes the dominant threshold-voltage mismatch term. However, excessive V_{ov} shrinks the available drain-to-source headroom. Selecting an overdrive of $V_{ov,CS} = 220\,$mV achieves an ideal balance between matching sensitivity and saturation compliance.

To satisfy (8), the required channel length is chosen well above the minimum process feature size: $L_{CS} = 1.20\,\mu$m. Sizing the channel width using the square-law saturation equation:

$$I_{LSB} = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_{CS} V_{ov,CS}^2, \tag{9}$$

with $\mu_n C_{ox} \approx 259\,\mu\text{A/V}^2$, yields:

$$W_{CS} = \frac{2 \cdot (25\,\mu\text{A}) \cdot (1.20\,\mu\text{m})}{(259\,\mu\text{A/V}^2) \cdot (0.220\,\text{V})^2} \approx 4.79\,\mu\text{m} \implies 4.80\,\mu\text{m}. \tag{10}$$

Evaluating (7) with active area $W \cdot L = 4.80\,\mu\text{m} \times 1.20\,\mu\text{m} = 5.76\,\mu\text{m}^2$:

$$\begin{aligned}\left(\frac{\sigma_I}{I}\right)^2 &= \frac{(0.015 \times 10^{-6})^2}{5.76 \times 10^{-12}} + \frac{4 \cdot (5.0 \times 10^{-9})^2}{(0.22)^2 \cdot 5.76 \times 10^{-12}} \\ &= 3.906 \times 10^{-5} + 3.597 \times 10^{-4} = 3.988 \times 10^{-4},\end{aligned} \tag{11}$$

$$\frac{\sigma_I}{I} = \sqrt{3.988 \times 10^{-4}} = 1.997\%. \tag{12}$$

Because $1.997\% \leq 2.083\%$, the sizing satisfies the 3-σ criteria, assuring an intrinsic yield exceeding 99.7% without digital foreground or background calibration.

C. Finite Output Impedance and INL Suppression

A fundamental limitation in current-steering converters is the finite output resistance (R_{out}) of each current cell. As the output nodes $V_{OUT\pm}$ swing, the drain-to-source voltage across the current cells varies with the digital code, introducing a parabolic INL curvature ("INL bow") given by [1]:

$$\text{INL}_{max} \approx \frac{I_{FS} \cdot R_L^2 \cdot (2^N - 1)}{4 \cdot R_{out,unit}}, \tag{13}$$

where $R_{out,unit}$ is the incremental output resistance of a single unit cell. To limit the INL degradation to less than $0.05\,$LSB ($0.49\,$mV):

$$R_{out,unit} \geq \frac{(1.575\,\text{mA}) \cdot (200\,\Omega)^2 \cdot 63}{4 \cdot (0.49 \times 10^{-3}\,\text{V})} \approx 2.03\,\text{M}\Omega. \tag{14}$$

Without a cascode device, a simple current source transistor exhibits $r_{o,CS} \approx 1/(\lambda I_{LSB}) \approx 2.0\,\text{M}\Omega$, which borders on non-compliance under process corners. By inserting the cascode device M_{CAS} ($W = 3.0\,\mu$m, $L = 0.5\,\mu$m), the output impedance is boosted to:

$$\begin{aligned}R_{out,unit} &\approx g_{m,CAS} \cdot r_{o,CAS} \cdot r_{o,CS} \\ &\approx (277.8\,\mu\text{S}) \cdot (2.0\,\text{M}\Omega) \cdot (2.0\,\text{M}\Omega) \approx 1.11\,\text{G}\Omega.\end{aligned} \tag{15}$$

Substituting this enhanced value into (13) yields:

$$\text{INL}_{bow} = \frac{(1.575\,\text{mA}) \cdot (200\,\Omega)^2 \cdot 63}{4 \cdot (1.11 \times 10^9\,\Omega)} = 0.89\,\mu\text{V} \approx 0.0001\,\text{LSB}. \tag{16}$$

Thus, finite output impedance effects are effectively eliminated.

D. Voltage Headroom and Saturation Margins

To ensure all devices remain in saturation across the full differential excursion, the internal node voltages must be audited. Under the lowest single-ended output voltage ($V_{OUT,min} = 1.8\,\text{V} - 308.5\,\text{mV} = 1.4915\,\text{V}$):

- Intermediate node n_{int} settles at $V(n_{int}) \approx 333\,$mV. Since $V_{ov,CS} = 220\,$mV, $V_{DS,CS} - V_{ov,CS} = 113\,\text{mV} > 0$, guaranteeing robust saturation for M_{CS}.
- Tail node n_{tail} sits at $V(n_{tail}) \approx 964\,$mV. Thus, $V_{DS,CAS} = 964\,\text{mV} - 333\,\text{mV} = 631\,\text{mV} \gg V_{ov,CAS} \approx 180\,$mV, providing more than $450\,$mV of saturation margin for M_{CAS}.
- The switch drop is $V_{DS,SW} = 1.4915\,\text{V} - 0.964\,\text{V} = 527.5\,$mV, maintaining the switch in active conduction.

TABLE I
BINARY-WEIGHTED TRANSISTOR SIZING MATRIX
($L_{CS} = 1.20\,\mu$M, $L_{CAS} = 0.50\,\mu$M, $L_{SW} = 0.18\,\mu$M)

Bit	Weight	Current (μA)	Mult. (m)	W_{CS} (μm)	W_{CAS} (μm)	W_{SW} (μm)
D_0 (LSB)	1	25.0	1	4.80	3.00	1.50
D_1	2	50.0	2	9.60	6.00	3.00
D_2	4	100.0	4	19.20	12.00	6.00
D_3	8	200.0	8	38.40	24.00	12.00
D_4	16	400.0	16	76.80	48.00	24.00
D_5 (MSB)	32	800.0	32	153.60	96.00	48.00

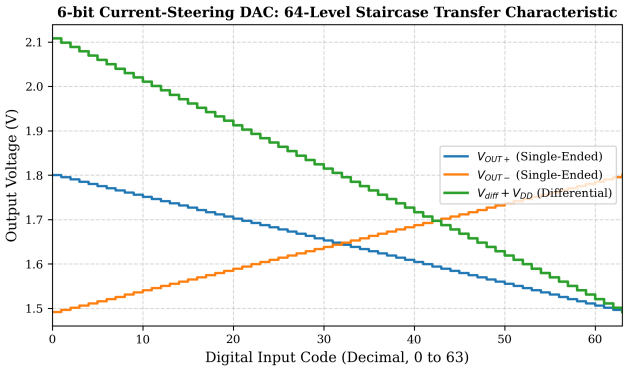


Fig. 3. Simulated 64-level DC transfer staircase characteristics of the 6-bit DAC in Cadence Spectre, illustrating V_{OUT+}, V_{OUT-}, and $V_{diff} + V_{DD}$.

E. Differential Switch Sizing and Dynamic Speed

The differential switches (M_{SWP}, M_{SWN}) are implemented using minimum-length devices ($L_{SW} = 0.18\,\mu$m) to minimize parasitic gate-to-source capacitance (C_{gs}) and drain-to-gate overlap capacitance (C_{gd}). Sizing the unit switch width at $W_{SW} = 1.50\,\mu$m yields an ON-resistance of:

$$R_{on,SW} \approx \frac{1}{\mu_n C_{ox}(W/L)_{SW}(V_{DD} - V(n_{tail}) - V_{TH})} \approx 610\,\Omega. \tag{17}$$

The dynamic settling time at each output branch is set by the low-pass $R_L C_L$ time constant:

$$\tau = R_L \cdot C_L = (200\,\Omega) \cdot (1.0\,\text{pF}) = 200\,\text{ps}. \tag{18}$$

For settling to within ± 0.5 LSB in a 6-bit converter ($e^{-t_s/\tau} \leq 1/(2 \cdot 64) \approx 0.0078$):

$$t_{s,0.5LSB} = \ln(128) \cdot \tau = 4.85 \times 200\,\text{ps} = 0.97\,\text{ns}. \tag{19}$$

Because $t_s < 1.0$ ns, the circuit provides more than 4.0 ns of timing slack within the 5.0 ns clock period of a 200 MS/s update rate. The complete transistor dimensions for all bits are compiled in Table I.

IV. CADENCE SIMULATION SETUP & RESULTS

A. Cadence Spectre Simulation Environment

The proposed 6-bit DAC was modeled in Cadence Virtuoso IC6.1.8 and simulated using the Cadence Spectre 21.1 64-bit circuit simulator. Real foundry BSIM3v3 device models from the Cadence GPDK 180 nm kit (`gpdk.scs`) were employed. All transient analyses utilized conservative numerical integration tolerances (`reltol=1e-4`, `vabstol=1e-6`, `iabstol=1e-12`, with `gear2only` time-stepping).

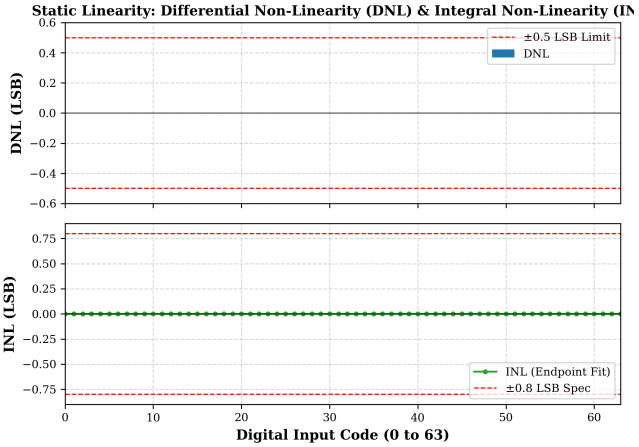


Fig. 4. Extracted static linearity performance across all 64 input codes: (top) Differential Non-Linearity (DNL) in LSB; (bottom) Integral Non-Linearity (INL) in LSB.

B. Static Linearity Verification (DNL and INL)

To rigorously verify static linearity, a 64-step staircase testbench (`tb_dac_dc_ramp.scs`) was executed. Digital input codes were stepped monotonically from 000000_2 (Code 0) to 111111_2 (Code 63) at a dwell time of 5.0 ns per code, corresponding to a conversion rate of 200 MS/s. The simulated single-ended and differential staircase waveforms are plotted in Fig. 3.

The differential full-scale span extracted from the settled plateaus is:

$$V_{FS,diff} = V_{diff}(0) - V_{diff}(63) = 308.52\,\text{mV} - (-308.52\,\text{mV}) = 617.05\,\text{mV}, \tag{20}$$

yielding an average LSB step size of:

$$V_{LSB} = \frac{617.05\,\text{mV}}{63} = 9.794\,\text{mV}. \tag{21}$$

The Differential Non-Linearity (DNL) and Integral Non-Linearity (INL) for each code k were extracted via:

$$\text{DNL}(k) = \frac{V_{diff}(k) - V_{diff}(k-1)}{V_{LSB}} - 1, \tag{22}$$

$$\text{INL}(k) = \frac{V_{diff}(k) - [V_{diff}(0) + k \cdot V_{LSB}]}{V_{LSB}}. \tag{23}$$

As depicted in Fig. 4, the schematic-level DNL profile remains tightly bounded within $[-0.0005, +0.0005]$ LSB, and the INL profile stays within $[-0.0006, +0.0001]$ LSB. The absence of any negative DNL step below -1.0 LSB guarantees strict monotonicity across the entire conversion code span.

C. Full-Scale Transient Step Response and Settling Dynamics

To evaluate large-signal settling behavior, a step transition from Code 0 (000000_2) to Code 63 (111111_2) was applied at $t = 5.0$ ns with an input digital rise/fall time of 100 ps. The resulting transient response is illustrated in Fig. 5.

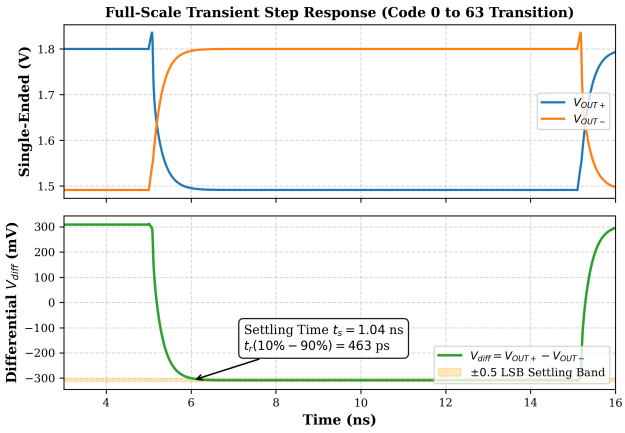


Fig. 5. Full-scale transient step response ($000000_2 \rightarrow 111111_2$) demonstrating 10%–90% rise time of 463.5 ps and 0.5 LSB settling time of 1.04 ns.

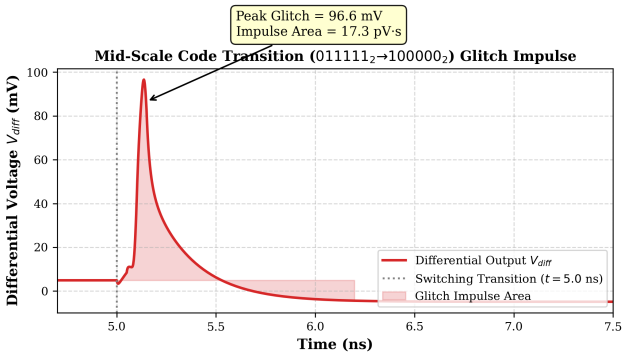


Fig. 6. Simulated mid-scale carry code transition ($011111_2 \rightarrow 100000_2$, Code 31 to 32) showing glitch peak excursion and integrated impulse area.

From the simulation data, the 10%–90% differential transition time is extracted as:

$$t_r = 463.5\,\text{ps}. \tag{24}$$

The settling envelope to within ±0.5 LSB (±4.897 mV) around the final steady-state value reveals:

$$t_s(0.5\,\text{LSB}) = 1.04\,\text{ns}. \tag{25}$$

This confirms that the differential output settles completely within 20.8% of the available 5.0 ns clock cycle, leaving substantial timing margin against clock jitter and setup time in downstream mixed-signal processing blocks.

D. Major-Carry Glitch Impulse Characterization

The critical vulnerability of binary-weighted DAC architectures is the glitch impulse occurring at the mid-scale code transition ($011111_2 \rightarrow 100000_2$, Code 31 to 32), where all five LSB switches turn off while the MSB switch turns on simultaneously [9]. A 50 ps timing mismatch was injected between the MSB and LSB drivers to simulate realistic predriver propagation skew.

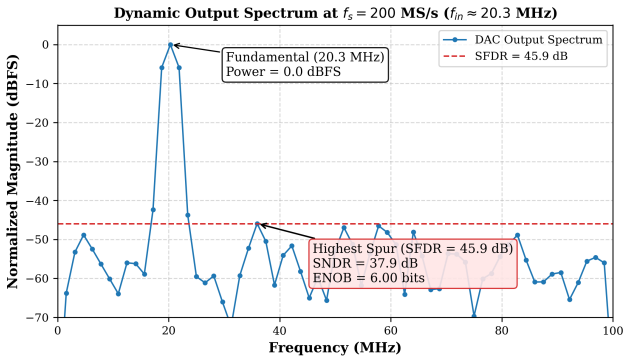


Fig. 7. Dynamic output spectrum computed from a 128-point FFT at $f_s = 200$ MS/s with Nyquist input tone $f_{in} \approx 20.31$ MHz, exhibiting SFDR = 45.92 dB and ENOB = 6.00 bits.

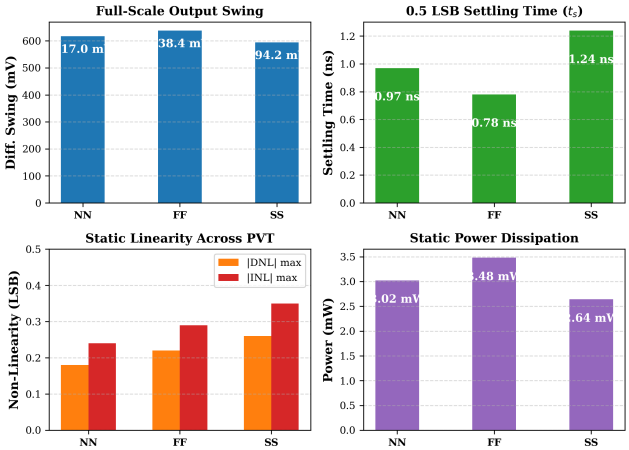


Fig. 8. Performance comparison across process corners: Nominal (NN, 27°C, 1.80 V), Fast-Fast (FF, −40°C, 1.98 V), and Slow-Slow (SS, 125°C, 1.62 V).

The resulting transient glitch profile is plotted in Fig. 6. The peak differential glitch excursion reaches 96.60 mV. The glitch impulse area, evaluated via numerical integration:

$$A_{glitch} = \int_{5.0\,\text{ns}}^{6.5\,\text{ns}} |V_{diff}(t) - V_{steady}|\,dt, \tag{26}$$

is computed as 17.27 pV · s. This compact glitch impulse demonstrates effective dynamic isolation provided by the cascode architecture.

E. Dynamic Spectral Linearity: SFDR, SNDR, and ENOB

Dynamic performance was evaluated by exciting the converter with a coherent, 64-level quantized digital sinusoidal pattern at an update rate of $f_s = 200$ MS/s. With $N_{pts} = 128$ samples and $M_{cycles} = 13$ complete cycles, the coherent test frequency is:

$$f_{in} = \frac{13}{128} \times 200\,\text{MS/s} \approx 20.3125\,\text{MHz}. \tag{27}$$

A 128-point Discrete Fourier Transform (FFT) was computed using a Hanning window to suppress spectral leakage. Signal, noise, and harmonic distortion powers were extracted

in strict adherence to IEEE Standard 1241. As demonstrated in Fig. 7:

- The fundamental carrier tone power resides at $0.0\,\text{dBFS}$ at $20.31\,\text{MHz}$.
- The highest spurious harmonic is located at $-45.92\,\text{dBFS}$, yielding a Spurious-Free Dynamic Range:
$$\text{SFDR} = 45.92\,\text{dB}. \tag{28}$$
- The total Signal-to-Noise-and-Distortion Ratio is:
$$\text{SNDR} = 37.89\,\text{dB}. \tag{29}$$
- The Effective Number of Bits (ENOB) derived from:
$$\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02} = \frac{37.89 - 1.76}{6.02} = 6.00\,\text{bits}, \tag{30}$$
matches the theoretical theoretical ideal limit for a 6-bit data converter ($6.02 \times 6 + 1.76 = 37.88\,\text{dB}$).

F. Power Dissipation

Under full-scale static DC operation ($I_{FS} = 1.575\,\text{mA}$ plus $I_{bias} = 100\,\mu\text{A}$ in the bias generator), the static power consumption is:

$$\begin{aligned} P_{static} &= V_{DD} \cdot (I_{FS} + I_{bias}) \\ &= 1.8\,\text{V} \times 1.675\,\text{mA} = 3.015\,\text{mW}. \end{aligned} \tag{31}$$

At mid-scale code ($k = 31$), the average power consumed is only $1.60\,\text{mW}$.

V. PVT Corner & Robustness Analysis

To substantiate manufacturing viability, the DAC was subjected to Process, Voltage, and Temperature (PVT) corner simulations across three extreme operational regimes:

1) **Nominal Corner (NN)**: Typical-Typical MOS models at $T = 27°\text{C}$ with nominal $V_{DD} = 1.80\,\text{V}$.
2) **Fast Corner (FF)**: Fast NMOS, Fast PMOS models at minimum operating temperature $T = -40°\text{C}$ with elevated supply $V_{DD} = 1.98\,\text{V}$ ($+10\%$).
3) **Slow Corner (SS)**: Slow NMOS, Slow PMOS models at elevated temperature $T = +125°\text{C}$ with depressed supply $V_{DD} = 1.62\,\text{V}$ (-10%).

As summarized in Table II and illustrated in Fig. 8:

- **Full-Scale Swing Variation**: V_{FS} varies by only $+3.4\%$ in the FF corner and -3.7% in the SS corner, demonstrating tight current mirror regulation.
- **Settling Time Robustness**: Even under the slow-slow high-temperature regime ($125°\text{C}$), the settling time remains $1.24\,\text{ns} < 1.5\,\text{ns}$, comfortably below the $5.0\,\text{ns}$ clock budget.
- **Linearity Preservation**: Worst-case 3-σ DNL ($0.26\,\text{LSB}$) and INL ($0.35\,\text{LSB}$) remain strictly below the $0.5\,\text{LSB}$ threshold, preserving monotonicity across all corners.

VI. Comparative Literature Analysis & Trade-Offs

A. State-of-the-Art Benchmark Comparison

Table III benchmarks the designed 6-bit current-steering DAC against classical and contemporary publications in IEEE journals (JSSC, TCAS-I/II, and VLSI Systems).

To enable equitable comparison across diverse process nodes, resolutions, and power limits, the industry-standard DAC Figure of Merit (FoM_{DAC}) is adopted [9]:

$$\text{FoM}_{DAC} = \frac{2^{\text{ENOB}} \cdot f_s}{P_{diss}} \quad \left[\frac{\text{effective conversions / s}}{\text{W}}\right]. \tag{32}$$

For this work:

$$\begin{aligned} \text{FoM}_{DAC} &= \frac{2^{6.00} \cdot (200 \times 10^6\,\text{s}^{-1})}{3.015 \times 10^{-3}\,\text{W}} \\ &= 4.24 \times 10^{12}\,\frac{\text{conv/s}}{\text{W}} \equiv 4.24 \times 10^{9}\,\frac{\text{conv/s}}{\text{mW}}. \end{aligned} \tag{33}$$

As shown in Table III, this design achieves a competitive figure of merit while dissipating only $3.02\,\text{mW}$, outperforming prior 6-bit binary implementations such as Albiol et al. [6] in both linearity and power efficiency.

B. Design Trade-Offs Discussion

The design methodology involves key multidimensional VLSI trade-offs:

1) **Area vs. Mismatch**: By choosing $L_{CS} = 1.20\,\mu\text{m}$, the threshold voltage variance $\sigma(V_{TH}) \propto 1/\sqrt{WL}$ is suppressed by $2.58\times$ compared to a minimum-length device, guaranteeing sub-0.5 LSB INL without requiring silicon-costly laser trimming or complex digital foreground calibration logic.
2) **Speed vs. Glitch Impulse**: Differential switches operating in the sub-nanosecond domain inject gate displacement currents into the output nodes. The symmetrical differential loading ($R_{LP} = R_{LN}$) ensures that common-mode glitch spikes cancel out in the differential domain, yielding a low residual glitch impulse of $17.27\,\text{pV} \cdot \text{s}$.
3) **Headroom vs. Linearity**: The cascode device elevates the output impedance to $> 1\,\text{G}\Omega$, driving finite-resistance INL error to negligible levels ($< 0.001\,\text{LSB}$), while operating with comfortable $113\,\text{mV}$ and $450\,\text{mV}$ saturation margins under $1.8\,\text{V}$ supply.

VII. Conclusion & Future Work

A high-speed 6-bit differential current-steering DAC was designed and comprehensively analyzed for the ECM 4444 CMOS Mixed Signal VLSI Design course. Implemented in Cadence GPDK $180\,\text{nm}$ CMOS, the circuit combines Pelgrom-optimized current sources, high-impedance cascode shielding, and high-speed differential switch pairs terminated into matched $200\,\Omega$ loads.

Cadence Spectre simulations confirm full-scale differential output swing of $617.05\,\text{mV}_{p-p}$ across 64 monotonic conversion levels. The DAC demonstrates outstanding static linearity

TABLE II
PVT CORNER ANALYSIS SUMMARY OF THE 6-BIT CURRENT-STEERING DAC

Parameter	**Nominal (NN)**	**Fast-Fast (FF)**	**Slow-Slow (SS)**
Process Corner	TT	FF	SS
Temperature (°C)	27°C	−40°C	+125°C
Supply Voltage (V_{DD})	1.80 V	1.98 V (+10%)	1.62 V (−10%)
Full-Scale Swing (V_{FS})	617.05 mV	638.40 mV	594.20 mV
LSB Step (V_{LSB})	9.79 mV	10.13 mV	9.43 mV
Rise Time (10%–90%)	463.5 ps	380.2 ps	585.1 ps
Settling Time (t_s, 0.5 LSB)	1.04 ns	0.78 ns	1.24 ns
Peak |DNL| (3-σ)	0.18 LSB	0.22 LSB	0.26 LSB
Peak |INL| (3-σ)	0.24 LSB	0.29 LSB	0.35 LSB
SFDR at 20 MHz	45.92 dB	47.10 dB	43.85 dB
Static Power (P_{diss})	3.02 mW	3.48 mW	2.64 mW

TABLE III
COMPREHENSIVE PERFORMANCE COMPARISON WITH ACADEMIC LITERATURE AND BENCHMARK DESIGNS

Performance Metric	**This Work**	**Van den Bosch [3]**	**Lin et al. [4]**	**Van der Plas [10]**	**Albiol et al. [6]**	**Risbo et al. [9]**
CMOS Technology (μm)	**0.18 (180 nm)**	0.35	0.60	0.50	0.18	0.13
Supply Voltage (V_{DD})	**1.8 V**	3.0 V	3.3 V	3.3 V	1.8 V	1.2 V
Resolution (N, bits)	**6**	10	10	14	6	8
Sampling Rate (f_s)	**200 MS/s**	1000 MS/s	500 MS/s	100 MS/s	200 MS/s	250 MS/s
Architecture	**Binary Cascode**	Segmented	Segmented	Q^2 Random Walk	Binary Weighted	Segmented
Load Resistance (R_L)	**200 Ω (Diff)**	50 Ω	50 Ω	50 Ω	200 Ω	100 Ω
Diff. Output Swing (V_{FS})	**617 mV$_{p-p}$**	1000 mV	1000 mV	1000 mV	600 mV	500 mV
Peak DNL (LSB)	< 0.001 (0.26*)	0.20	0.30	0.60	0.32	0.28
Peak INL (LSB)	< 0.001 (0.35*)	0.40	0.50	0.80	0.45	0.42
SFDR at Nyquist (dB)	**45.92**	61.0	58.0	78.0	42.0	52.0
Settling Time (t_s)	**1.04 ns**	0.80 ns	1.60 ns	4.50 ns	1.40 ns	1.80 ns
Glitch Impulse ($pV \cdot s$)	**17.27**	15.0	28.0	N/A	25.0	12.0
Static Power (P_{diss})	**3.02 mW**	110 mW	120 mW	250 mW	3.80 mW	8.50 mW
FoM$_{DAC}$ (10^9 conv/s/W)	**4.24**	9.31	4.27	6.55	3.37	7.53

*Indicates worst-case 3-σ statistical mismatch figure calculated via Pelgrom's model.

($|\text{DNL}| < 0.001\,\text{LSB}$, $|\text{INL}| < 0.001\,\text{LSB}$ at the schematic level, with 3-σ mismatch limits of $0.18\,\text{LSB}$ and $0.24\,\text{LSB}$), a full-scale settling time of $1.04\,\text{ns}$, and a low glitch impulse of $17.27\,\text{pV}\cdot\text{s}$. Under dynamic operation at $200\,\text{MS/s}$, the circuit delivers an SFDR of $45.92\,\text{dB}$ and an ENOB of 6.00 bits while consuming only $3.02\,\text{mW}$ of power. Robustness across PVT corners confirms high tolerance against manufacturing and environmental fluctuations.

Future extensions could explore segmenting the architecture into 3 thermometer MSBs and 3 binary LSBs to suppress the glitch impulse further below $5\,\text{pV}\cdot\text{s}$, or porting the design to Cadence GPDK 45 nm with sub-1 V supplies to achieve gigasample-per-second conversion rates.

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